Example: A μ- meson with an average lifetime $2 \times 10^{-6} s$ is created in the upper atmosphere at an elevation of 6000 meter. When is created it has velocity 0.998 c in the direction towards the earth. What is the average distance it will travel before decaying as determine by an observer on earth. Also explain the situation for an observer sitting on the μ- meson frame

Classically, the distance travel before it decay is $2 \times 10^{-6} \times (0.998 \times 3 \times 10^{8}) = 599\ m$

(unable to reach the earth surface)

Relativistic correction : Time measure in a moving frame has been delayed. To an observer at earth lifetime of the μ- meson would be $= \gamma T = \frac{1}{\sqrt{1-(0.998)^2}} \times 2 \times 10^{-6} = 31.6 \times 10^{6} s$

So the distance travel before it decay will be = $31.6 \times 10^{-6} \times (0.998 \times 3 \times 10^{8}) = 9470\ m$

(can be easily reach the earth surface)

To an observer at μ- meson, the distance 6000 m (measure by an earth observer) will appear to be contracted distance and it has to travel only $= \mathrm{L}/\gamma = \sqrt{1-(0.998)^2} \times 6000 = 379\ m$

(can be easily manged)

Relativistic Velocity Addition

$$x' = \gamma(x - vt)$$
$$y' = y$$
$$z' = z$$
$$t' = \gamma\left(t - \frac{vx}{c^2}\right)$$

$$dx' = \gamma(dx - v\,dt)$$
$$dy' = dy$$
$$dz' = dz$$
$$dt' = \gamma\left(dt - \frac{v\,dx}{c^2}\right)$$

$$u_x{}' = \frac{dx'}{dt'} = \frac{\gamma(dx - v\,dt)}{\gamma\left(dt - \frac{v\,dx}{c^2}\right)} = \frac{\frac{dx}{dt} - v}{1 - \frac{v}{c^2}\frac{dx}{dt}}$$

$$u_x{}' = \frac{u_x - v}{1 - \frac{vu_x}{c^2}}$$

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$$u_y{}' = \frac{dy'}{dt'} = \frac{dy}{\gamma\left(dt - \frac{v\,dx}{c^2}\right)} = \frac{\frac{dy}{dt}}{\gamma\left(1 - \frac{v}{c^2}\frac{dx}{dt}\right)}$$

$$u_y{}' = \frac{u_y}{\gamma\left(1 - \frac{vu_x}{c^2}\right)}$$

Relativistic Velocity Addition

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$\Rightarrow$

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$$u_y' = \frac{u_y}{\gamma\left(1 - \frac{vu_x}{c^2}\right)}$$

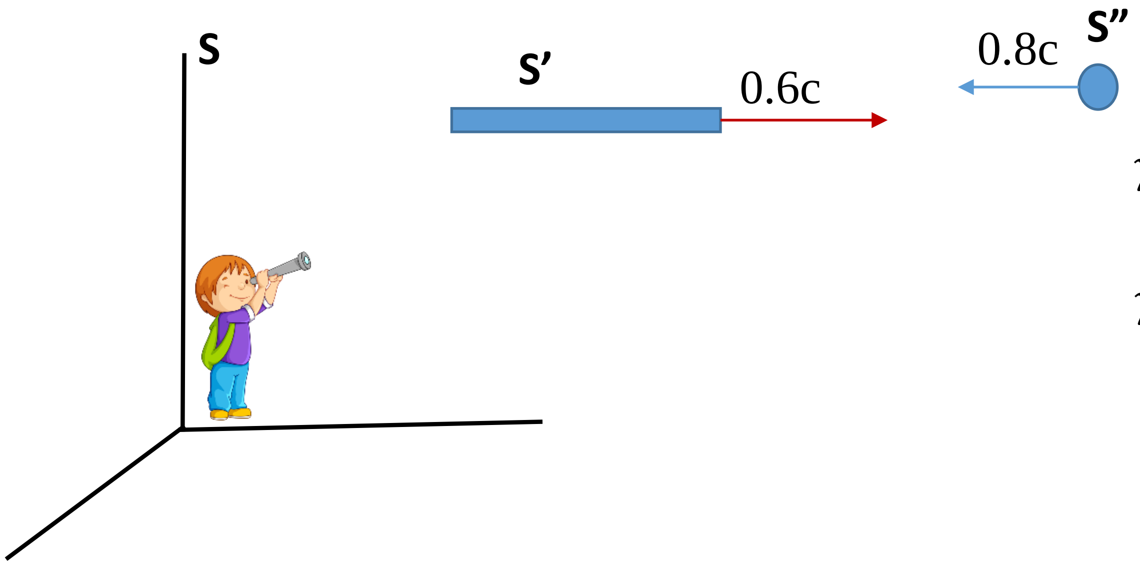
$$u_z' = \frac{u_z}{\gamma\left(1 - \frac{vu_x}{c^2}\right)}$$

If $u_x = \mathrm{c}$ & $u_y = u_z = 0$ then

$$u_x' = \frac{c-v}{1-\frac{vc}{c^2}} = c, \;\& \; u'_y = u'_z = 0$$

Speed of light remain constant in all inertial frame of reference

Example: An observer at S(earth) frame observes one rod of unit length moving with velocity 0.6c along the +ve x direction and a point particle is moving along –ve x direction with a velocity 0.8c. How much time the particle will take to cross the rod to an observer when s/he sitting on the rod-frame, on the particle frame and on the earth ?



$$\gamma_{SS'} = \frac{1}{\sqrt{1-(0.6)^2}} = \frac{5}{4}$$

$$\gamma_{SS''} = \frac{1}{\sqrt{1-(0.8)^2}} = \frac{5}{3}$$

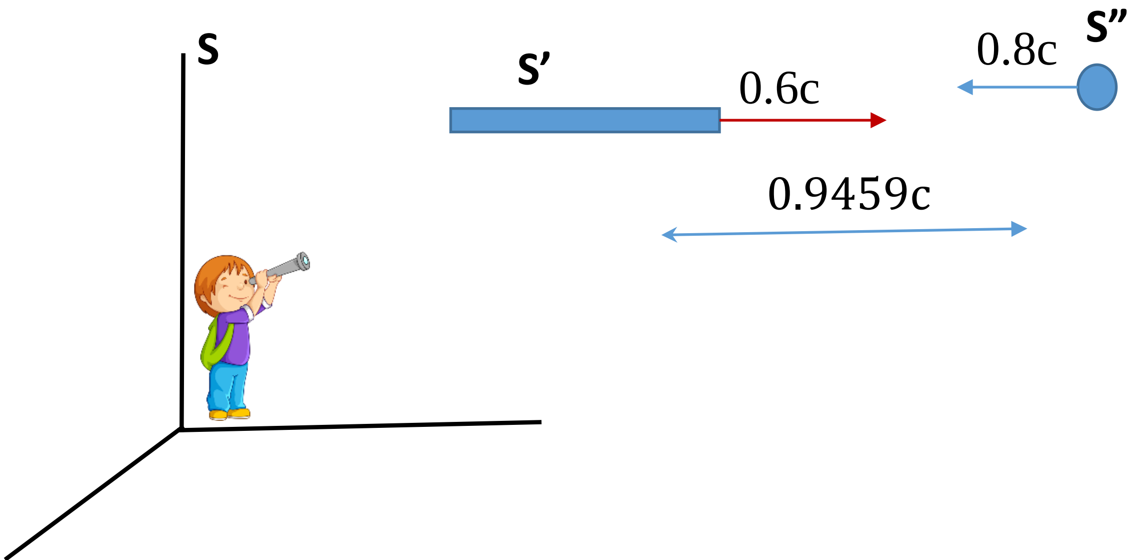
To the observer at S'-frame, the length is proper and he can measure the time of crossing. For that he needs to calculate the velocity of the particle with respect to him.

Using velocity addition rule: $u_x{}' = \frac{u_x - v}{1-\frac{v u_x}{c^2}} = \frac{-0.8c-0.6c}{(1-0.6\times(-0.8))} = 0.9459\text{c}$

The observer at S'-frame will see that the particle is approaching towards the rod with a velocity 0.9459c and it will take $T_{S'}$ =1/0.9459c sec = $3.52 \times 10^{-9} sec$.

Example: An observer at S(earth) frame observes one rod of unit length moving with velocity 0.6c along the +ve x direction and a point particle is moving along –ve x direction with a velocity 0.8c. How much time the particle will take to cross the rod to an observer when s/he sitting on the rod-frame, on the particle frame and on the earth ?

$T_{S'} = 3.52 \times 10^{-9} sec$



$$\gamma_{SS'} = \frac{1}{\sqrt{1-(0.6)^2}} = \frac{5}{4}$$

$$\gamma_{SS''} = \frac{1}{\sqrt{1-(0.8)^2}} = \frac{5}{3}$$

To the observer at S''-frame, the time is proper and he can measure the time considering first end and second end of the rod crossing him. However, to this observer the 1 meter rod is approaching him with a velocity 0.9459c. Therefore, the length of the rod will be contracted

$$\gamma_{S'S''} = \frac{1}{\sqrt{1-(0.9459)^2}} = 3.082$$

So the contracted length will be $\frac{1}{\gamma_{S'S''}} = 0.3245m,$

And the observed time of crossing will be $\boldsymbol{T_{S''} = \frac{0.3245}{0.9459c} = 1.1434 \times 10^{-9} s}$

Example: An observer at S(earth) frame observes one rod of unit length moving with velocity 0.6c along the +ve x direction and a point particle is moving along –ve x direction with a velocity 0.8c. How much time the particle will take to cross the rod to an observer when s/he sitting on the rod-frame, on the particle frame and on the earth ?

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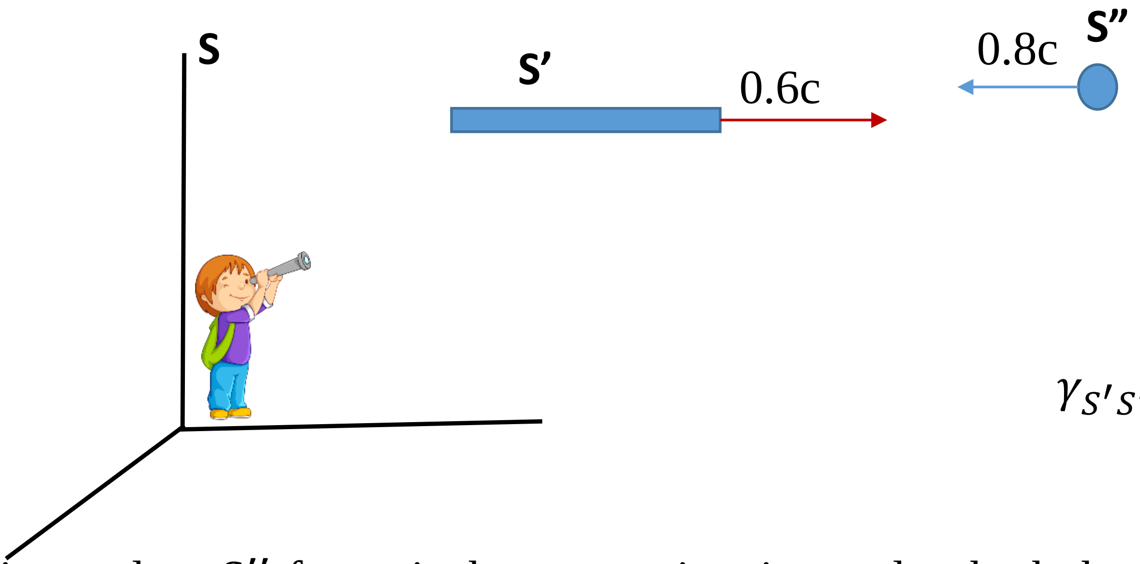

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$$\gamma_{S'S''} = \frac{1}{\sqrt{1-(0.9459)^2}} = 3.082$$

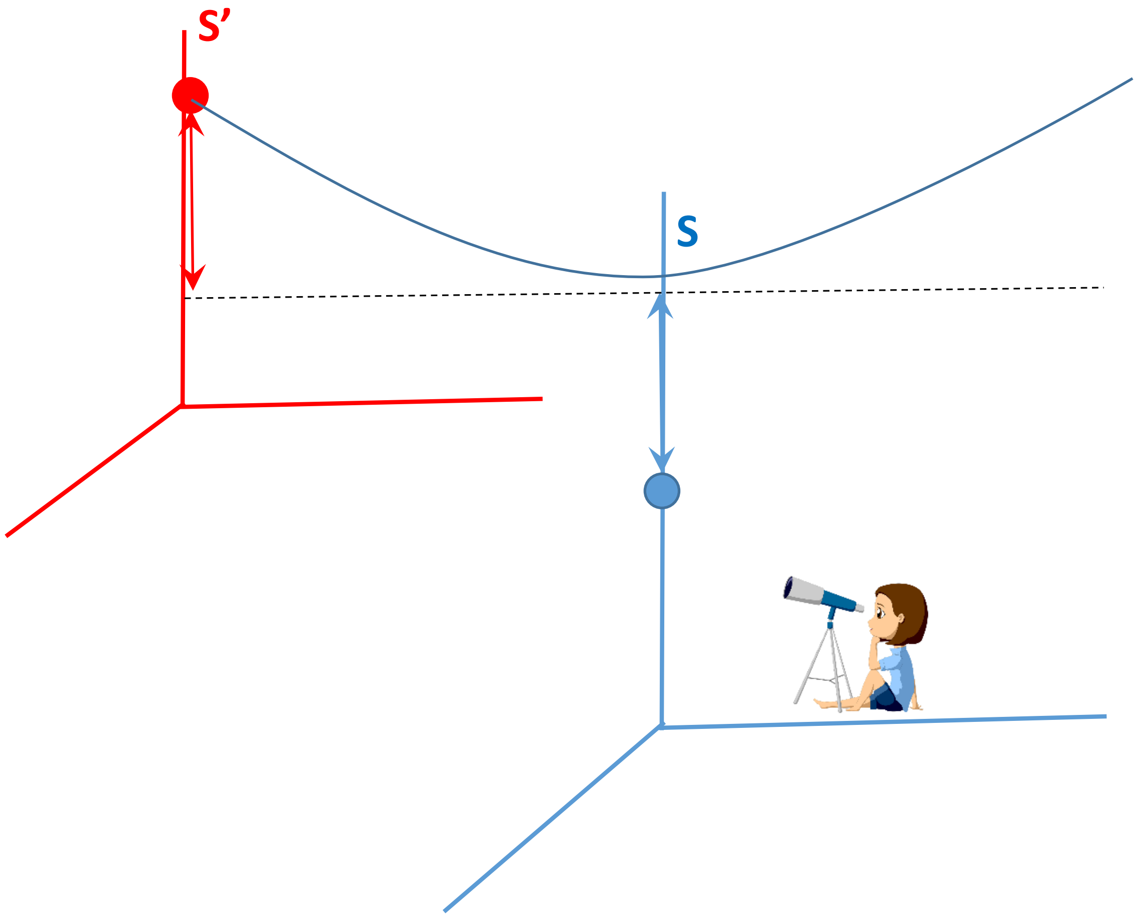
Now the measured time interval at S''-frame is the proper time interval as both the event is measured in the same place. So we can use this time interval to estimate the time interval estimated by the observer at S frame.

To the observer at S frame, time $T_{S''}$ is delayed by $\gamma_{SS''}$. Therefore to him, the time taken by the particle to cross the rod will be $T_s = \gamma_{SS''} \times T_{S''} = \frac{5}{3} \times 1.1434 \times 10^{-9} s = 1.9056 \times 10^{-9} s$

Relativistic momentum

Relativistic momentum

Two identical particle A of mass m_A in S-frame, and particle B of mass m'_B in S'-frame moving with velocity V_A and V'_B moving along +ve and –ve Y direction respectively in their respective frame, so that $V_A = V'_B$ & $m_A = m'_B$. They travel l distance, elastically collided and return back their original position in their respective frame.



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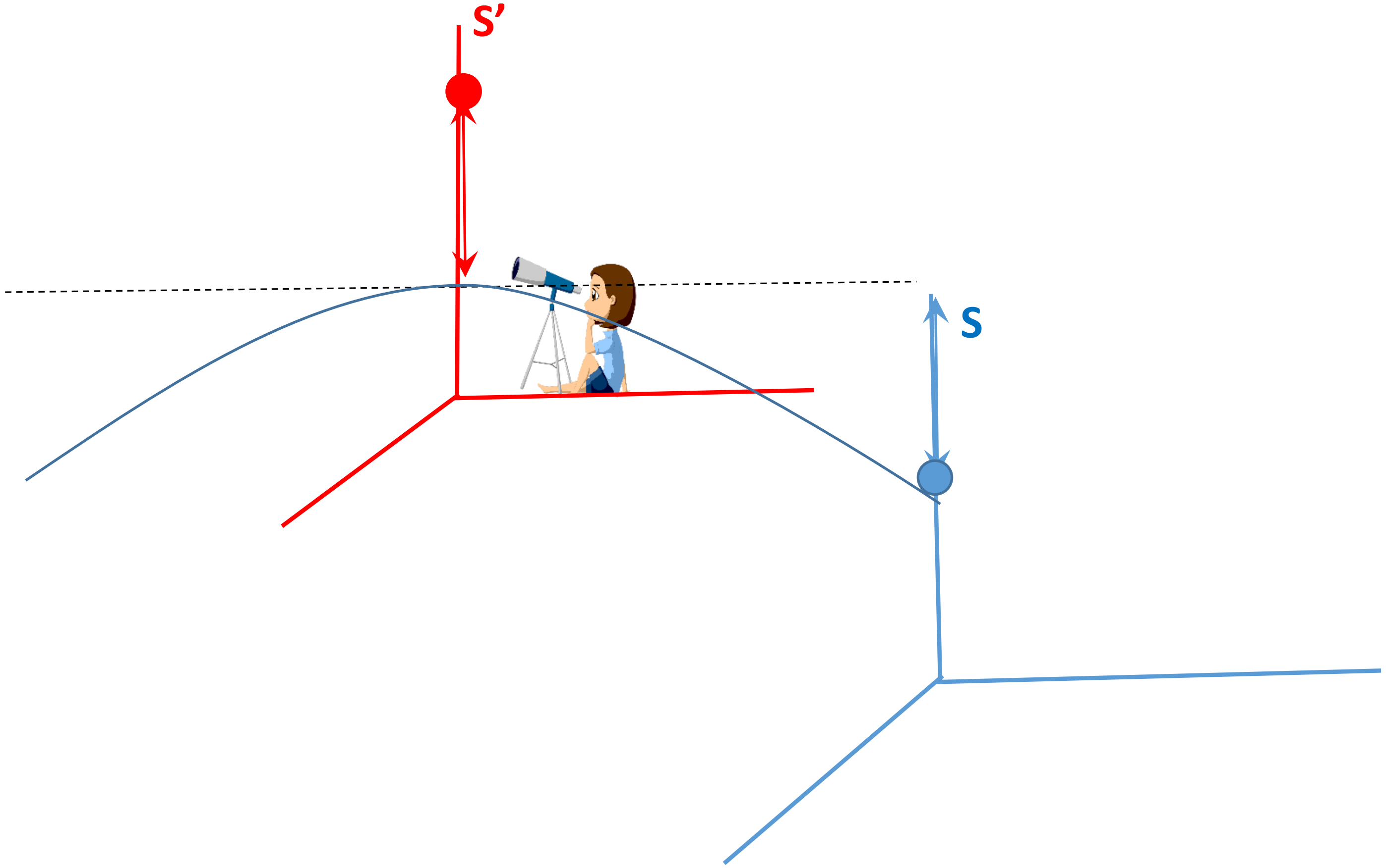
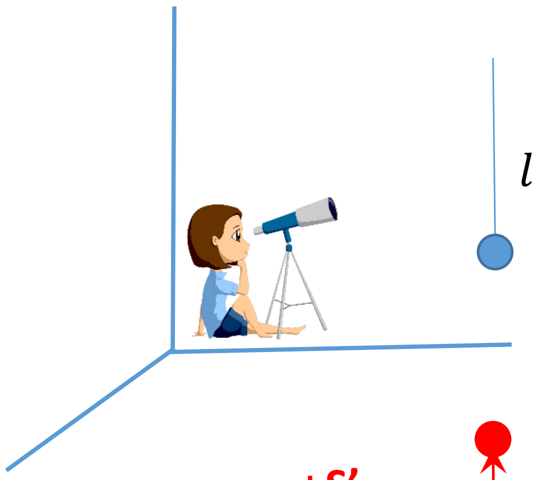
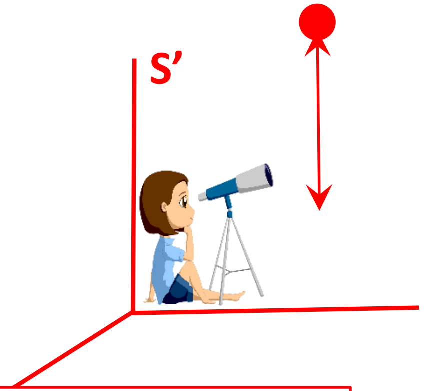


To observer at S frame, time take by the particle A to travel l distance up collide and return back to its original position is $T_S^A = \frac{2l}{V_A}$



To observer at S' frame, time take by the particle B to travel l distance down collide and return back to its original position is $T_{S'}^B = \frac{2l}{V_B'}$



What will be the measured time for the particle B to the observer at S ???

To the observer at S frame $T_{S'}^B$ will be delayed by γ $\quad T_S^B = \gamma T_{S'}^B = \gamma \frac{2l}{V_B'}$

To the observer at S frame, velocity of the B particle is $V_B = \frac{2l}{T_S^B} = \frac{V_B'}{\gamma}$ and the momentum would be $p_B = m_B \frac{V_B'}{\gamma}$, Where m_B is the mass of B particle to the S frame observer.

This momentum p_B should be the same as the momentum measured by the observer at S' −frame

ie. $p'_B = m'_B V_B' = m_B \frac{V_B'}{\gamma}$

$$p'_B = m'_B V'_B = m_B \frac{V'_B}{\gamma}$$

$$\boxed{m_B = \gamma m'_B}$$

In the limit of $V'_B = 0$, and the particle B sitting at the origin of the S'-frame, the observer at S frame will realize that a particle of mass $m_B = \gamma m'_B = \gamma m_A$, whose rest mass is m_A, approaching to him with a velocity v (the velocity of the S' frame). So the momentum of the particle will be $p = \gamma m_A v$

So a particle of mass m moving with a velocity v has momentum

$$p = \gamma m v = \frac{mv}{\sqrt{1-\left(\frac{v}{c}\right)^2}}$$

Relativistic Acceleration

Particle of mass m moving with a velocity v has momentum $p = \gamma m v = \dfrac{mv}{\sqrt{1-\left(\frac{v}{c}\right)^2}}$

$$\text{Force } \vec{F} = \frac{d\vec{p}}{dt} = \left[\frac{m}{\sqrt{1-\left(\frac{v}{c}\right)^2}} - mv \times \left(-\frac{1}{2}\right)\frac{1}{\left(1-\left(\frac{v}{c}\right)^2\right)^{3/2}} \times \frac{2v}{c^2}\right] \times \frac{d\vec{v}}{dt}$$

$$= \quad m \times \left[\frac{1}{\sqrt{1-\left(\frac{v}{c}\right)^2}} + \frac{1}{\left(1-\left(\frac{v}{c}\right)^2\right)^{3/2}} \times \frac{v^2}{c^2}\right] \times \frac{d\vec{v}}{dt}$$

$$= \frac{m}{\sqrt{1-\left(\frac{v}{c}\right)^2}} \times \left[\frac{1-\frac{v^2}{c^2}+\frac{v^2}{c^2}}{\left(1-\left(\frac{v}{c}\right)^2\right)}\right] \times \vec{a}$$

$$= \frac{m}{\left(1-\left(\frac{v}{c}\right)^2\right)^{3/2}} \times a$$

$$\boxed{\vec{a} = \frac{\vec{F}}{m} \times \left(1 - \left(\frac{v}{c}\right)^2\right)^{3/2}}$$

Relativistic Work-Energy relation

Work done by a constant force $\vec{F}$ acting on a body which is displaced a distance $\vec{s}$

$$W = \vec{F} \cdot \vec{s}$$

If the body start from rest to an observer, the work done is same as the KE

$$KE = \int_0^s \vec{F} \cdot d\vec{s} = \int_0^s \frac{d\vec{p}}{dt} \cdot d\vec{s} = \int_0^s \frac{d(\gamma m v)}{dt} ds = \int_0^s d(\gamma m v) \frac{ds}{dt} = \int_0^v v \, d(\gamma m v)$$

Integration by parts

$$z = 1 - \left(\frac{v}{c}\right)^2$$

$$dz = -\frac{2v}{c^2} dv$$

$$vdv = -\frac{c^2}{2} dz$$

$$KE = \left[\frac{v\, mv}{\sqrt{1-\left(\frac{v}{c}\right)^2}}\right]_0^v - \int_0^v \frac{mv \; dv}{\sqrt{1-\left(\frac{v}{c}\right)^2}}$$

$$= \frac{mv^2}{\sqrt{1-\left(\frac{v}{c}\right)^2}} + \frac{mc^2}{2} \int_1^{1-\left(\frac{v}{c}\right)^2} \frac{dz}{\sqrt{z}} = \gamma m v^2 + \frac{mc^2}{2} \times 2 \times \left[\sqrt{1-\left(\frac{v}{c}\right)^2} - 1\right] = \gamma m v^2 + \frac{mc^2}{\gamma} - mc^2$$

$$= (\gamma - 1) mc^2$$

Relativistic Work-Energy relation

$$KE = (\gamma - 1)mc^2$$

$$KE + mc^2 = \gamma mc^2 = Total\ Energy$$

When an object of mass m is at rest, then KE=0 but the total energy is $= mc^2 = rest\ energy$

$$\boxed{\mathrm{E} = mc^2}$$

Relativistic Energy-momentum relation

The relativistic momentum $p = \gamma m v$ and The relativistic total energy $\mathrm{E} = \gamma m c^2$

$$p^2 = (\gamma m v)^2 = \frac{m^2 v^2}{\left(1 - \left(\frac{v}{c}\right)^2\right)} = \frac{m^2 c^2 v^2}{(c^2 - v^2)}$$

$$p^2 c^2 = \frac{m^2 c^4 v^2}{(c^2 - v^2)} = \frac{m^2 c^4 (v^2 - c^2 + c^2)}{(c^2 - v^2)} = -m^2 c^4 + \frac{m^2 c^4}{(1 - v^2/c^2)}$$

$$p^2 c^2 + m^2 c^4 = \gamma^2 m^2 c^4 = (\gamma m c^2)^2 = E^2$$

$$\boxed{E^2 = p^2 c^2 + m^2 c^4}$$

Spacetime Metrics

A **metric** defines the distance between two points.

Example : Cartesian Co-ordinate system the distance between two point (x_1, y_1, z_1) and (x_2, y_2, z_2) is

$$ds^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2 \geq 0$$

In spacetime we can define an **event** as something marked by the 4 coordinates x, y, z, and t.
And a **metric**

$$ds^2=(c\ dt)^2 - (dx^2 + dy^2 + dz^2)$$

This metric has the advantage of being **invariant** under a Lorentz transformation

1. A body explode and have two parts, each of them has 1 kg mass, moving with 0.6c relative to the original body. What was the mass of original body?

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3. An electron has a kinetic energy of 0.1 MeV. Finds its speed according to classical and relativistic mechanics.

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4. A particle has a KE 20 times its rest energy. Find the speed of the particle in the unit of c.

6. Find the momentum of an electron whose speed is 0.6c in the unit of MeV/c. Given that the mass of an electron is 511 KeV/c^2

7. Find the momentum of an electron whose KE is same as its rest energy. Given the mass of an electron is 511 KeV/c^2

8. Find the speed and momentum (in GeV/c unit) of a proton, whose total energy is 3.5 GeV. Given that the proton has rest energy of 0.938 GeV.

9. A particle has a KE of 62 MeV, and momentum 335 MeV/c, Find its mass (in the unit of MeV/c^2) and speed in the unit of c.

Example: A space craft moving in -*ve* x direction receives a light signal from a source in *xy*-plane. In the reference frame of the fixed star, the speed of the space craft is v and the signal arrives at an angle θ to the axis of the space craft. With the help of Lorentz transformation find the angle θ' which the signal arrives in the reference frame of the space craft.